

Code-Layout, Readability and Reusability

Date	1 July 2026
Team ID	SWTID-2026-9068
Project Name	Smart Lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

The project follows Python coding standards with proper indentation, meaningful naming conventions, modular architecture, reusable components, and effective error handling. The code is readable, maintainable, and can be easily extended for future enhancements.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Used standard Python indentation (4 spaces) throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Project is organized into frontend, backend, services, models, routers, and reports folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like <code>event_text</code> , <code>fact_result</code> , <code>history_data</code> , <code>feedback</code> , and <code>report_path</code> clearly indicate their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as <code>analyze_event()</code> , <code>Verify_topic()</code> , and <code>generate_topics()</code> use meaningful names.
5	Code Comments	Inline and block comments explain logic	Yes	Important functions and API logic are documented using comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Backend logic is divided into reusable service modules and API routers.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable functions and modular architecture reduce code duplication.

8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-except blocks are used in API requests and frontend operations.
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Event Analyzer Service	Python (Transformers)	Event analysis and topic extraction	High
2	Fact Checker Service	Python (Wikipedia API)	Fact verification module	High
3	History Logger	Python (JSON)	Stores and retrieves conversation history	High
4	Feedback Logger	Python (JSON)	Stores user feedback from frontend	High
5	PDF Generator	Python (ReportLab)	Generates downloadable reports	Medium

Overall Code Quality Assessment:

S.No	Component / Module Name	Language / Technology
1	Event Analyzer Service	Python (Transformers)
2	Fact Checker Service	Python (Wikipedia API)
3	History Logger	Python (JSON)
4	Feedback Logger	Python (JSON)
5	PDF Generator	Python (ReportLab)